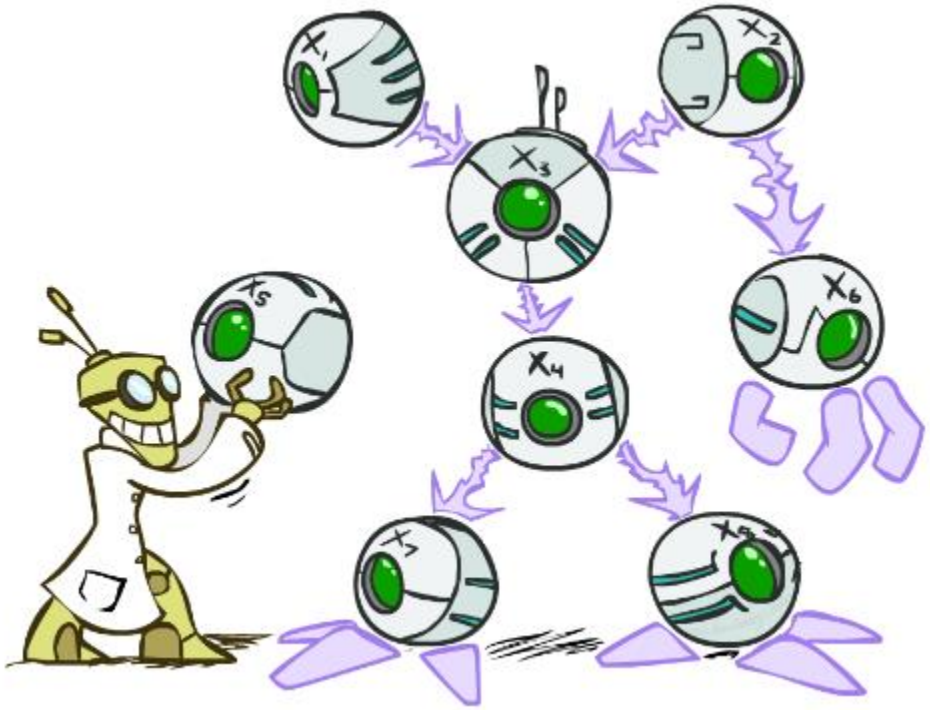
CS 188: Artificial Intelligence

Bayes' Nets



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[These slides were created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All CS188 materials are available at http://ai.berkeley.edu.]

Slides modified by Toon Calders

Motivating Example

- Acute Pancreatitis causes an overexpression of Lipase and Amylase (both are pancreatic enzymes).
- A Systemic Inflammation Reaction (SIR) may be caused by Acute Pancreatitis or Pneumonia.
- A SIR may cause an elevated white blood cell count and increased body temperature.
- Pneumonia and Asthma may both cause an increased Respitatory Rate and decreased PO2 (Partial Pressure of Oxygen).

Motivating Example

- *Our patient is a known Asthma patient, has an elevated blood cell count, an elevated level of amylase, but a normal body temperature. What is the probability that our patient has a systemic inflammation reaction?*

- How to model our knowledge?
- How to make inferences?



Probability: Notation

- Random variables: X, Y, Z
- Values: x, y, z
- Probability: P(X=x, Y=y, Z=z)
  - If no confusion: P(x, y, z)
- Conditional probability: P(X=x | Z=z)
- P(X | Y) = P(X) $\equiv$ $\forall x \forall y$: P(X=x, Y=y) = P(X=x)
- Joint probability distribution of X, Y, Z: P(X,Y,Z)
- Conditional probability distribution of X and Y, given Z: P(X,Y|Z)

Joint Probability Distribution

d_1	d_2	s	$P(d_1,d_2,s)$
1	1	1	0
1	1	2	1/36
1	1	3	0
…	…	…	…
1	1	12	0
1	2	1	0
1	2	2	0
1	2	3	1/36
…	…	…	…
6	6	10	0
6	6	11	0
6	6	12	1/36

$$
\begin{aligned}
P(D_1 = 5 \mid S = 11) \\
&= \frac{P(D_1 = 5, S = 11)}{P(S = 11)} \\
&= \frac{\sum_{d_2=1}^{6} P(D_1 = 5, D_2 = d_2, S = 11)}{\sum_{d_1=1}^{6} \sum_{d_2=1}^{6} P(D_1 = d_1, D_2 = d_2, S = 11)} \\
&= \frac{1/36}{1/18} \\
&= 0.5
\end{aligned}
$$



Probability Recap

- Conditional probability $$P(x|y) = \frac{P(x,y)}{P(y)}$$
- Product rule $$P(x,y) = P(x|y)P(y)$$
- Chain rule $$\begin{aligned} P(X_1, X_2, \ldots X_n) &= P(X_1)P(X_2|X_1)P(X_3|X_1, X_2)\ldots \\ &= \prod_{i=1}^{n} P(X_i|X_1, \ldots, X_{i-1}) \end{aligned}$$
- X, Y independent if and only if: $$\forall x, y : P(x,y) = P(x)P(y)$$
- X and Y are conditionally independent given Z if and only if:

$$\forall x, y, z : P(x,y|z) = P(x|z)P(y|z) \qquad X \perp\!\!\!\perp Y|Z$$

Example: Independence?

$P(T)$

T	P
hot	0.5
cold	0.5

$P_1(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$P_2(T, W)$

T	W	P
hot	sun	0.3
hot	rain	0.2
cold	sun	0.3
cold	rain	0.2

$P(W)$

W	P
sun	0.6
rain	0.4

Independence

- D1 : value of die 1
- D2 : value of die 2
- S : sum of dice

$$D_1 \perp\!\!\!\perp D_2$$

$$D_1 \not\perp\!\!\!\perp D_2 \mid S$$

$$S \not\perp\!\!\!\perp D_1$$



Example: Independence

- N fair, independent coin flips:

$P(X_1)$

H	0.5
T	0.5

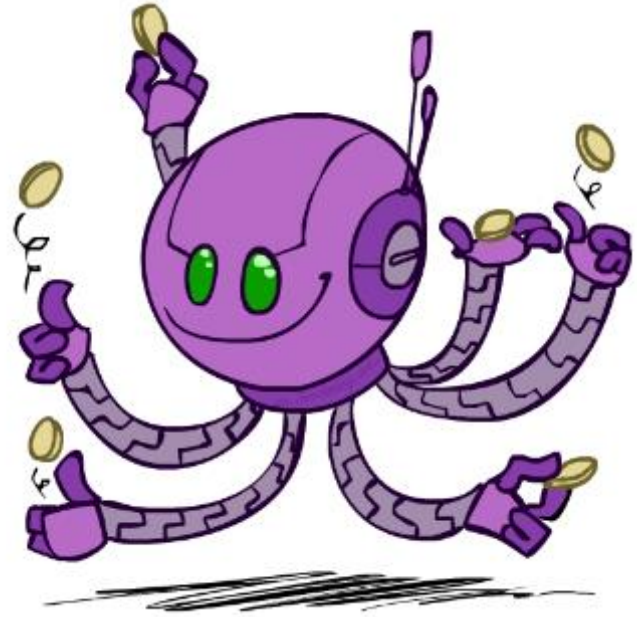
$P(X_2)$

H	0.5
T	0.5

$\ldots$

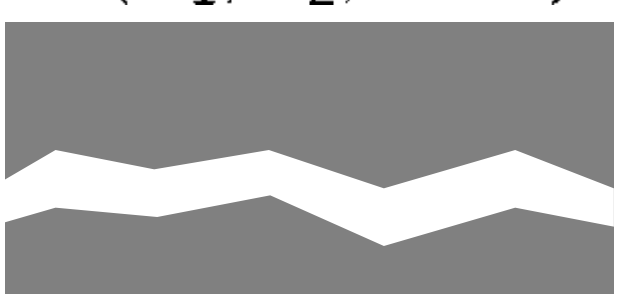
$P(X_n)$

H	0.5
T	0.5



$P(X_1, X_2, \ldots X_n)$

2^n





Conditional Independence

- *Conditional independence* is our most basic and robust form of knowledge about uncertain environments.
- X is conditionally independent of Y given Z $X \perp\!\!\!\perp Y | Z$

if and only if:

$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$

or, equivalently, if and only if

$$\forall x, y, z : P(x|z, y) = P(x|z)$$

Conditional Independence

- What about this domain:
  - Traffic
  - Umbrella
  - Raining



Conditional Independence and the Chain Rule

- Chain rule: $$P(X_1, X_2, \ldots X_n) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2)\ldots$$

- Trivial decomposition:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) =$$
$$P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain}, \text{Traffic})$$

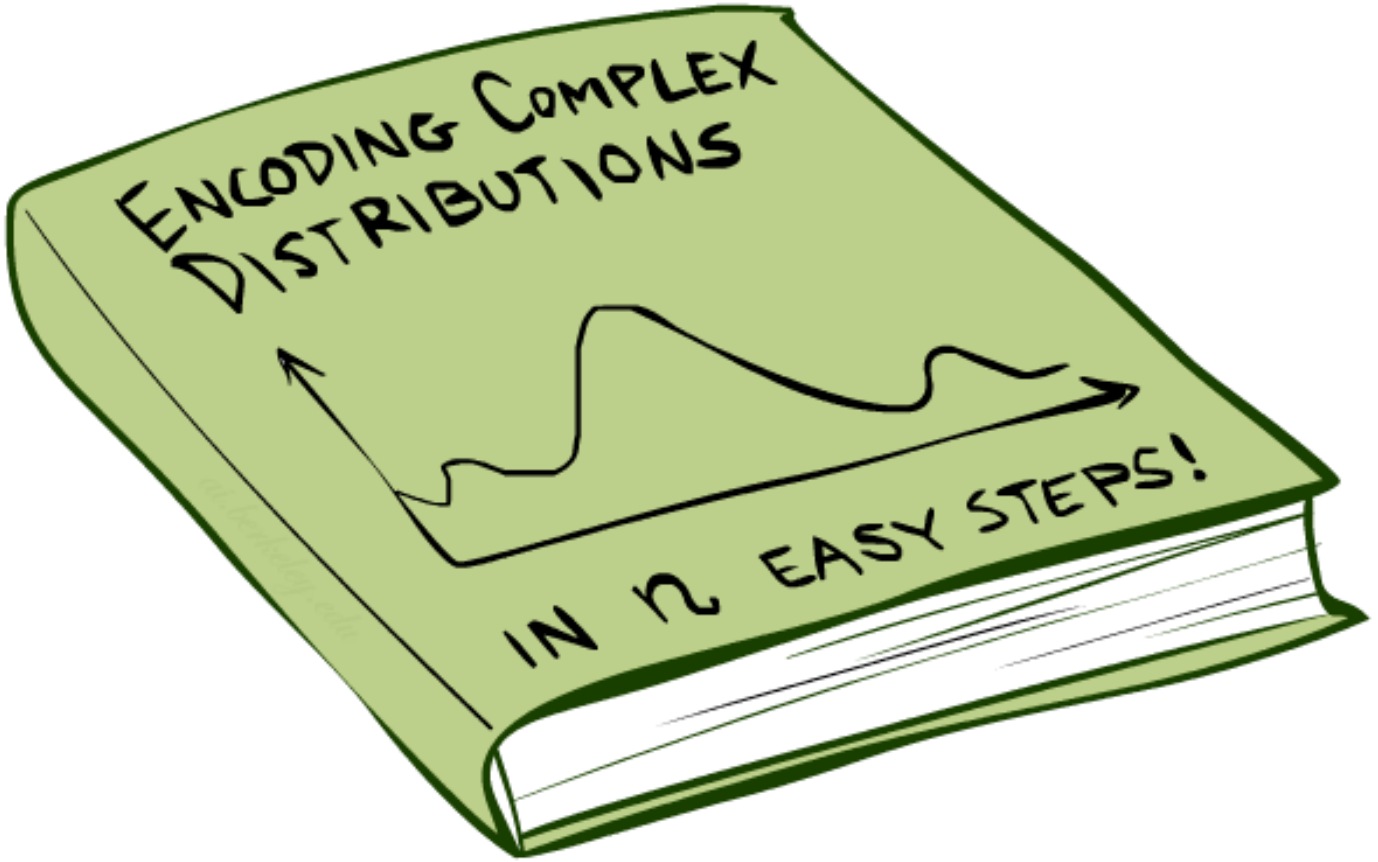


- With assumption of conditional independence:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) =$$
$$P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$$

- Bayes'nets / graphical models help us express conditional independence assumptions

Bayes'Nets: Big Picture

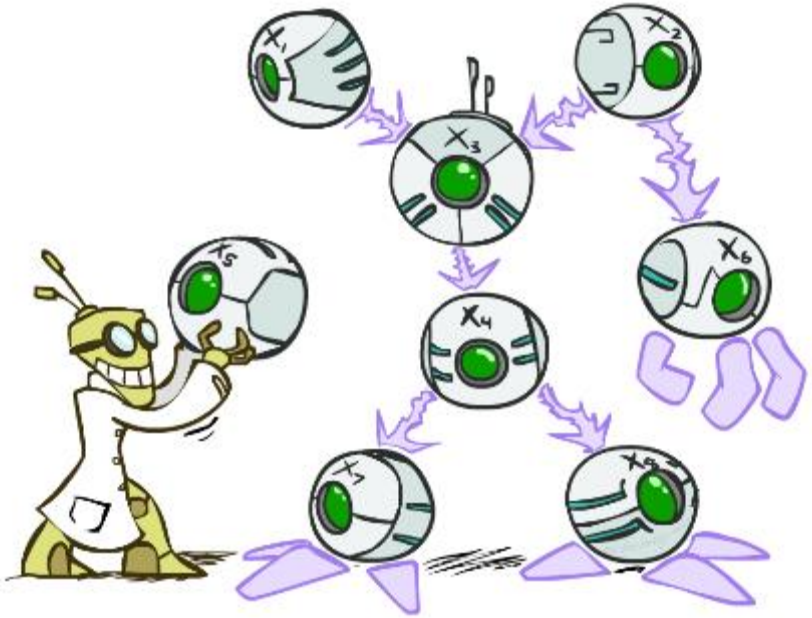


Bayes' Nets: Big Picture

- Two problems with using full joint distribution tables as our probabilistic models:
  - Unless there are only a few variables, the joint is WAY too big to represent explicitly
  - Hard to learn (estimate) anything empirically about more than a few variables at a time



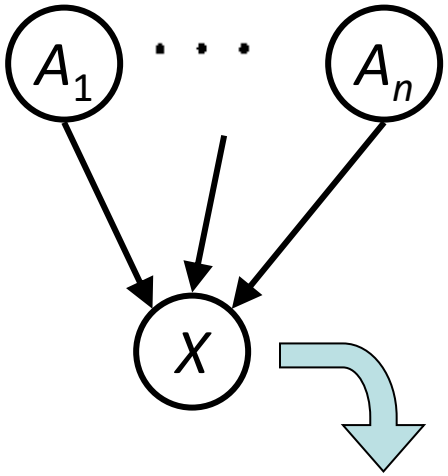
- Bayes' nets: a technique for describing complex joint distributions (models) using simple, local distributions (conditional probabilities)
  - More properly called graphical models
  - We describe how variables locally interact
  - Local interactions chain together to give global, indirect interactions



Bayes' Net Semantics

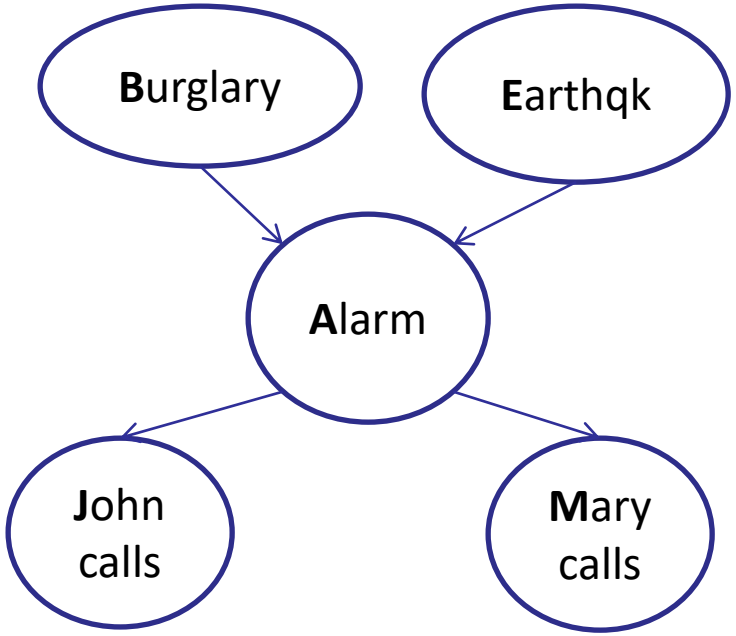


- A set of nodes, one per variable X
- A directed, acyclic graph
- Every node is conditionally independent from all its non-descendants given its parents



$$P(X|A_1 \ldots A_n)$$

Example: Alarm Network

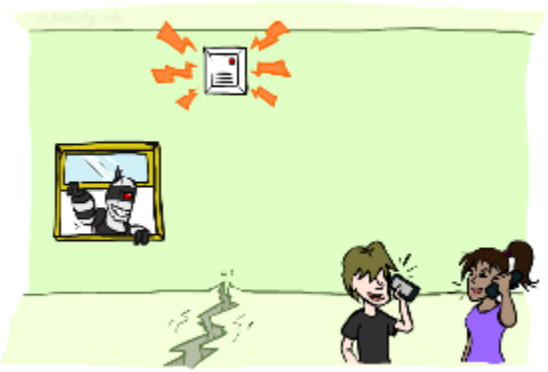


$$B \perp\!\!\!\perp E$$

$$E \perp\!\!\!\perp B$$

$$J \perp\!\!\!\perp B, E, M | A$$

$$M \perp\!\!\!\perp B, E, J | A$$



$$\begin{aligned} P(B, E, A, J, M) &= P(B)P(E|B)P(A|B, E)P(J|B, E, A)P(M|B, E, A, J) \\ &= P(B)P(E)P(A|B, E)P(J|A)P(M|A) \end{aligned}$$

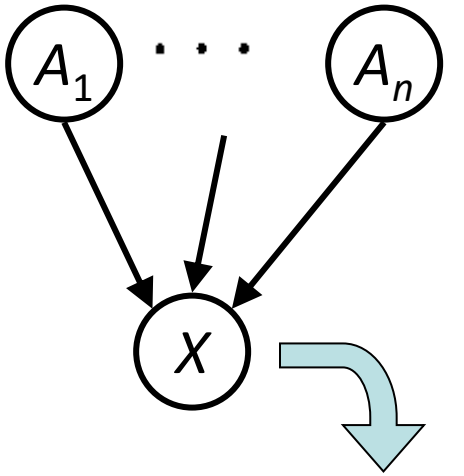
Bayes' Net Semantics



- A set of nodes, one per variable X
- A directed, acyclic graph
- Every node is conditionally independent from all its non-descendants given its parents
- This information allows for this decomposition:

$$P(X_1, \ldots, X_k) = \prod_{i=1}^{k} P(X_i \mid \text{Parents}(Xi))$$

- Hence, next to network structure, also store the distributions $P(X_i|\text{Parents}(X_i))$
  - E.g., Conditional probability tables



$P(X|A_1 \ldots A_n)$

A Bayes net = Topology (graph) + Local Conditional Probabilities

Size of a Bayes' Net

- How big is a joint distribution over N Boolean variables?

  2^N

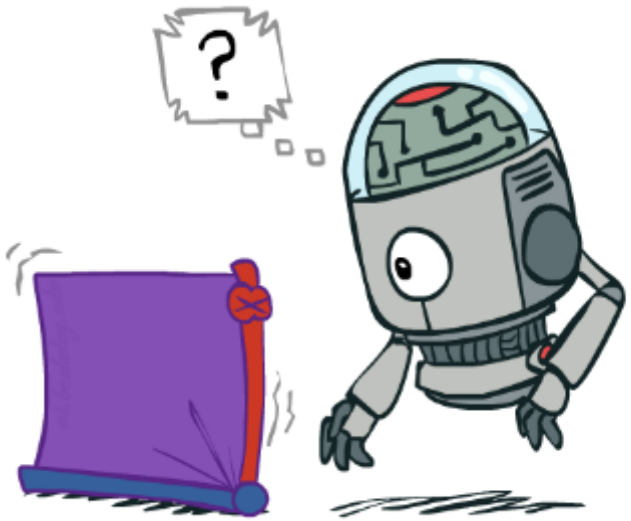
- How big is an N-node net if nodes have up to k parents?

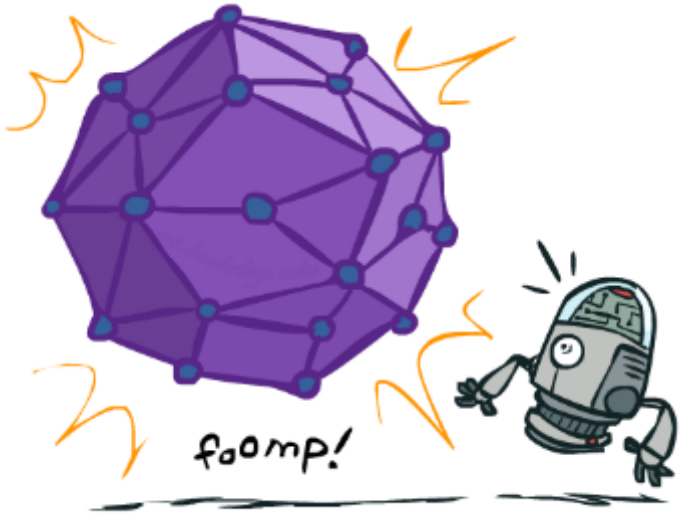
  $O(N * 2^{k+1})$

- Both give you the power to calculate

$$P(X_1, X_2, \ldots X_n)$$

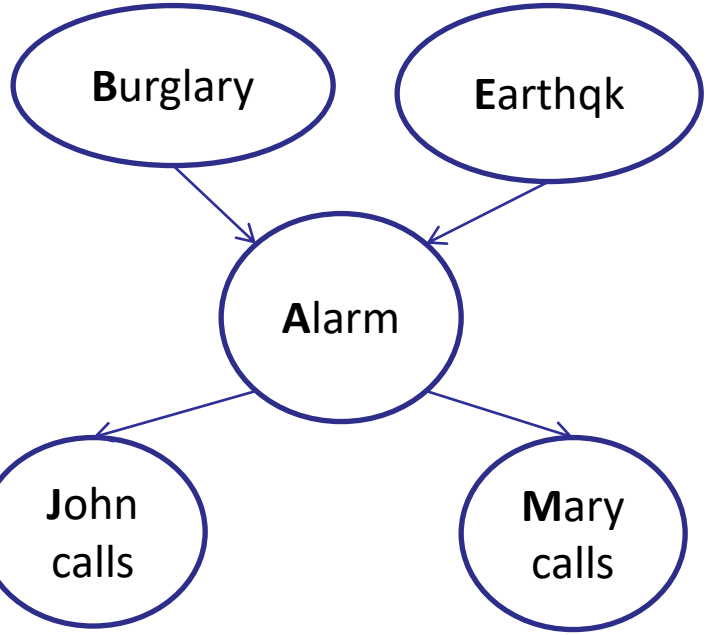
- BNs: Huge space savings!
- Also easier to elicit local CPTs
- Also faster to answer queries (coming)



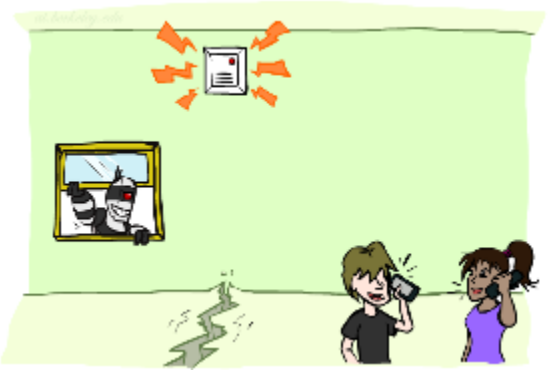


Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998



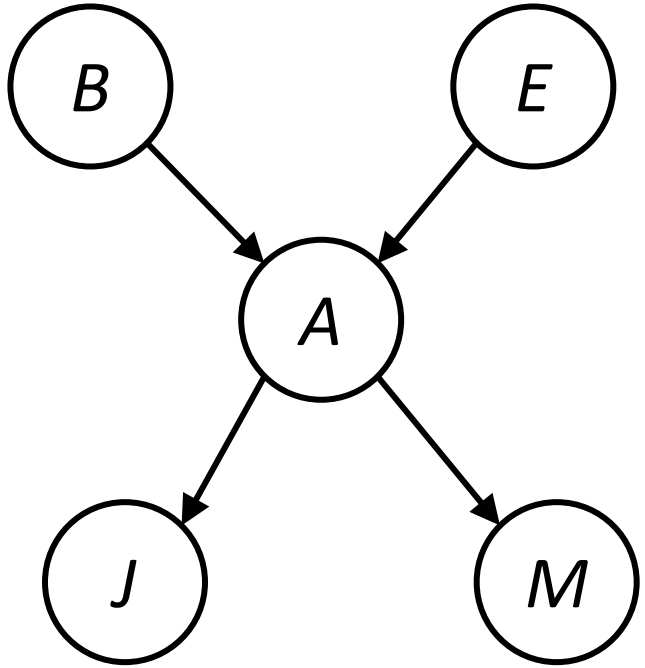
A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

Example: Alarm Network

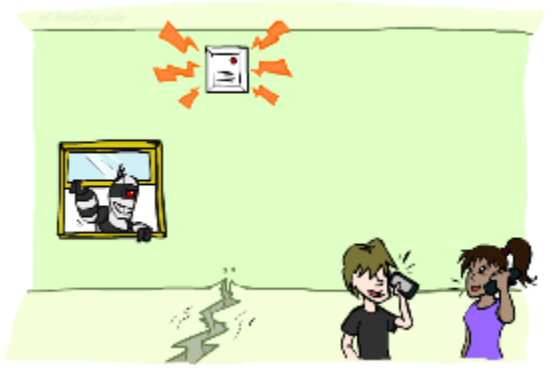
B	P(B)
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B	E	A	P(A|B,E)
+b	+e	+a	0.95
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+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

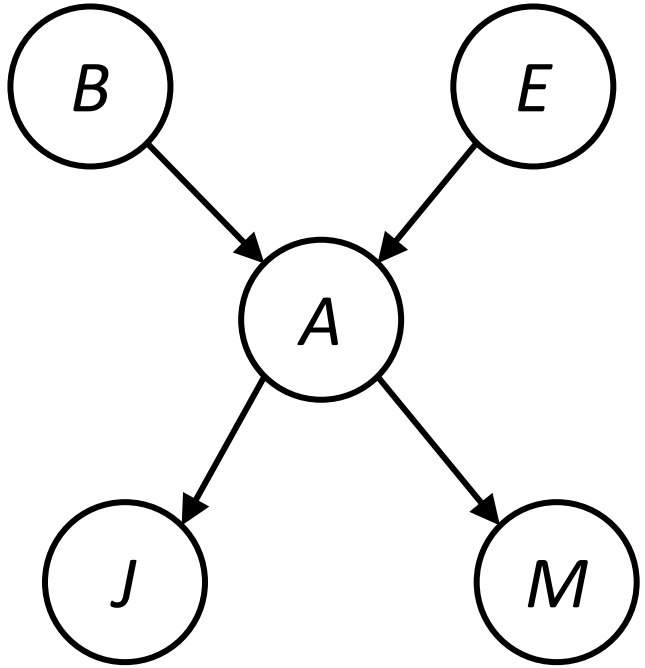
$$P(+b, -e, +a, -j, +m) =$$

$$= P(+b)P(-e)P(+a|+b,-e)P(-j|+a)P(+m|+a)$$

Example: Alarm Network

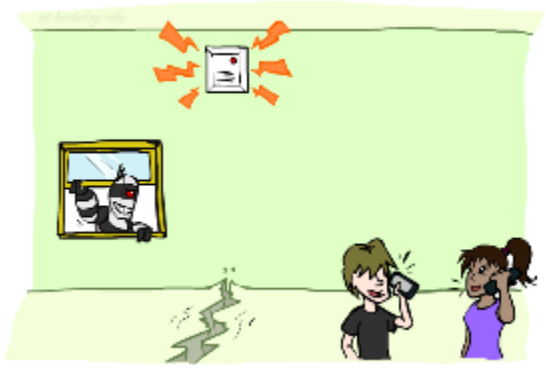
B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998



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-a	+j	0.05
-a	-j	0.95

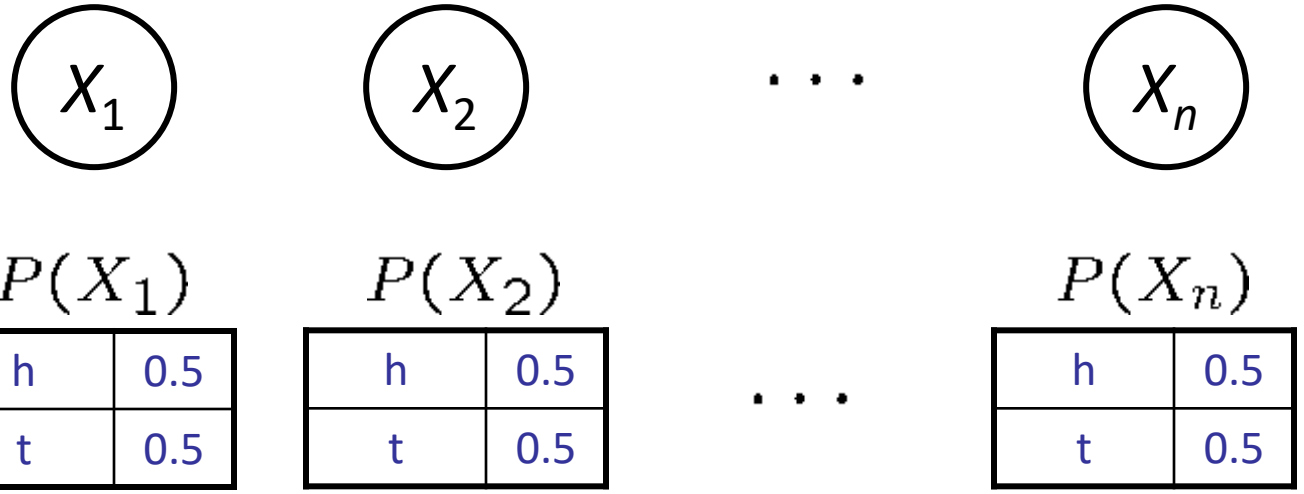
A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

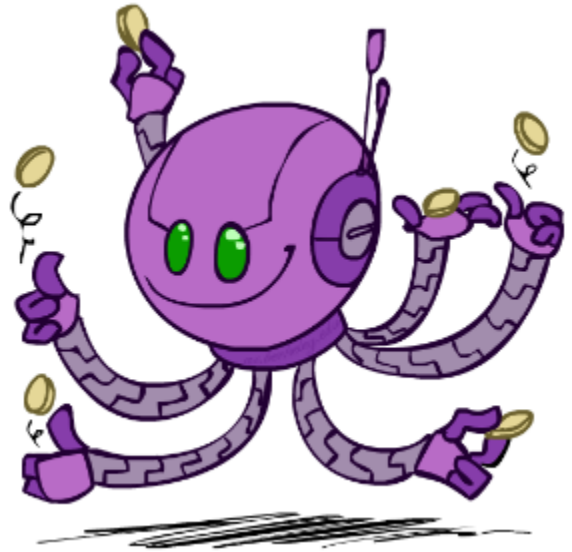


B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, -e, +a, -j, +m)$$
$$= P(+b)P(-e)P(+a|+b,-e)P(-j|+a)P(+m|+a)$$
$$= 0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7$$

Example: Coin Flips

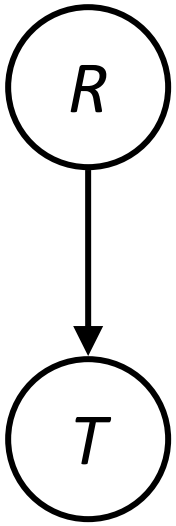




$$P(h, h, t, h) =$$

Only distributions whose variables are absolutely independent can be represented by a Bayes' net with no arcs.

Example: Traffic



$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

$$P(+r, -t) =$$



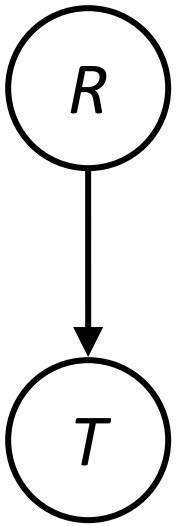


Example: Traffic

- Causal direction







$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

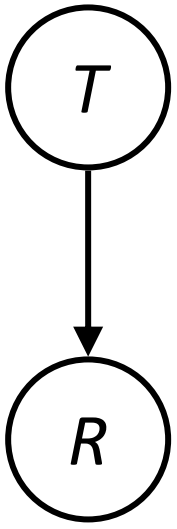
$P(T, R)$

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16

Example: Reverse Traffic

- Reverse causality?





$P(T)$

+t	9/16
-t	7/16

$P(R|T)$

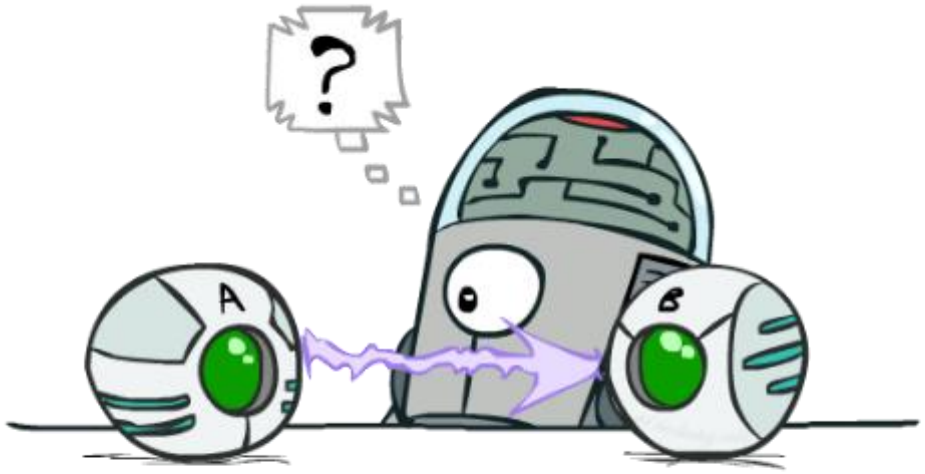
+t	+r	1/3
	-r	2/3
-t	+r	1/7
	-r	6/7

$P(T, R)$

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16

Causality?

- When Bayes' nets reflect the true causal patterns:
  - Often simpler (nodes have fewer parents)
  - Often easier to think about
  - Often easier to elicit from experts



- BNs need not actually be causal
  - Sometimes no causal net exists over the domain (especially if variables are missing)
  - E.g. consider the variables *Traffic* and *Drips*
  - End up with arrows that reflect correlation, not causation

- What do the arrows really mean?
  - Topology may happen to encode causal structure
  - Topology really encodes conditional independence

$$P(x_i|x_1, \ldots x_{i-1}) = P(x_i|parents(X_i))$$

Motivating Example

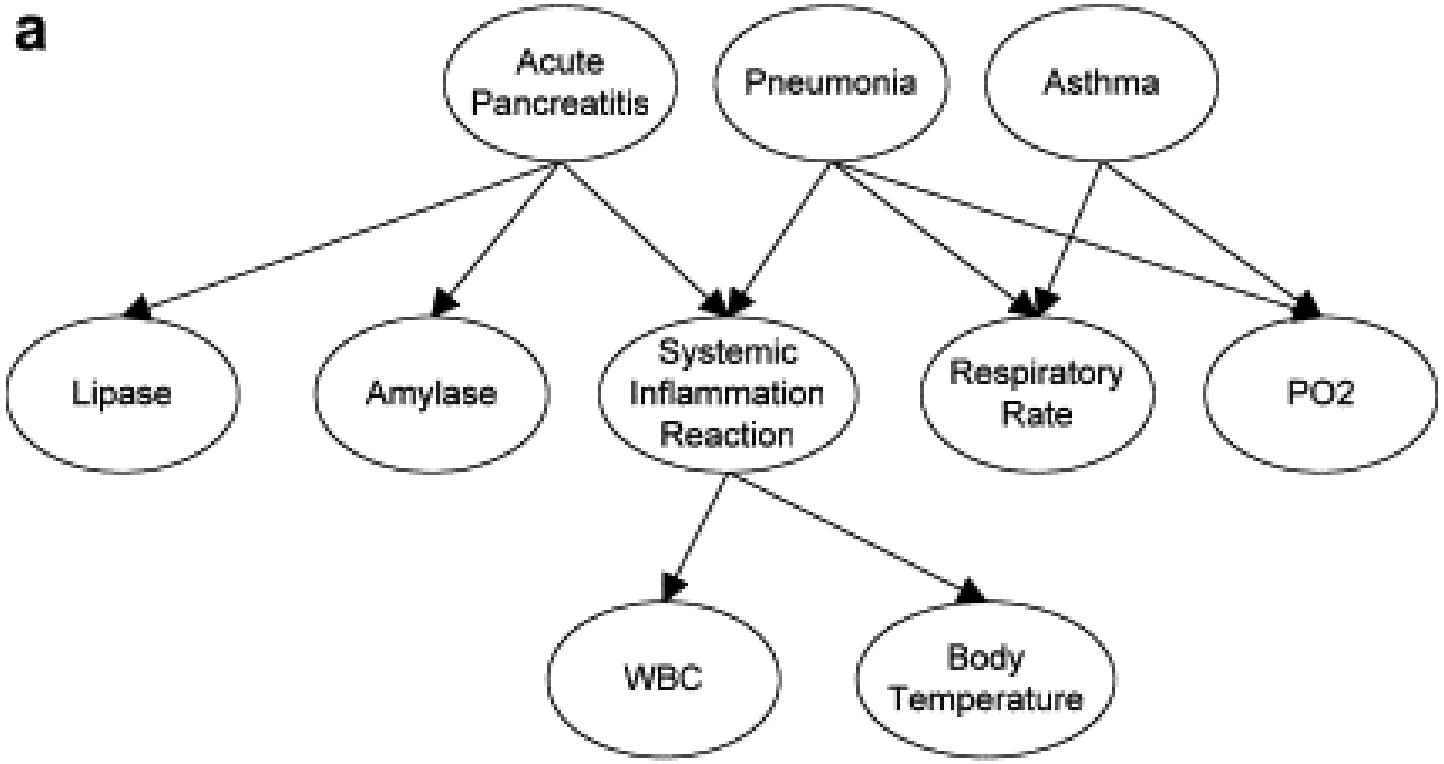
- *Our patient is a known Asthma patient, has an elevated blood cell count, an elevated level of amylase, but a normal body temperature. What is the probability that our patient has a systemic inflammation reaction?*

- How to model our knowledge?
- How to make inferences?



Motivating Example: Bayesian Net

a



b

Pneumonia	Asthma	PO2	Probability
Positive	Positive	<60	0.6
		60~80	0.2
		>80	0.2
Positive	Negative	<60	0.3
		60~80	0.3
		>80	0.4
Negative	Positive	<60	0.4
		60~80	0.3
		>80	0.3
Negative	Negative	<60	0.05
		60~80	0.1
		>80	0.85

Lin, Jau-Huei, and Peter J. Haug. "Exploiting missing clinical data in Bayesian network modeling for predicting medical problems." *Journal of biomedical informatics* 41.1 (2008): 1-14.

Side Note: Conditional Independence

- Every node is independent from its non-descendants given its parents
- Can we generalize this?

We will see the definition of D-separation in a minute

- *D-separation* generalizes this:

It is guaranteed that $X \perp\!\!\!\perp Y | Z$

if and only if

Every node in X is D-separated from every node in Y by Z

D-separation: Outline

- Study independence properties for triples
- Analyze complex cases in terms of member triples
- D-separation: a condition / algorithm for answering such queries

Causal Chains

- This configuration is a "causal chain"



X: Low pressure Y: Rain Z: Traffic

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

- Guaranteed X independent of Z ? *No!*
  - One example set of CPTs for which X is not independent of Z is sufficient to show this independence is not guaranteed.
  - Example:
    - Low pressure causes rain causes traffic, high pressure causes no rain causes no traffic
    - In numbers:

      P(+y | +x) = 1, P(-y | - x) = 1,
      P(+z | +y) = 1, P(-z | -y) = 1

Causal Chains

- This configuration is a "causal chain"



X: Low pressure Y: Rain Z: Traffic

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

- Guaranteed X independent of Z given Y?

$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$

$$= \frac{P(x)P(y|x)P(z|y)}{P(x)P(y|x)}$$

$$= P(z|y)$$

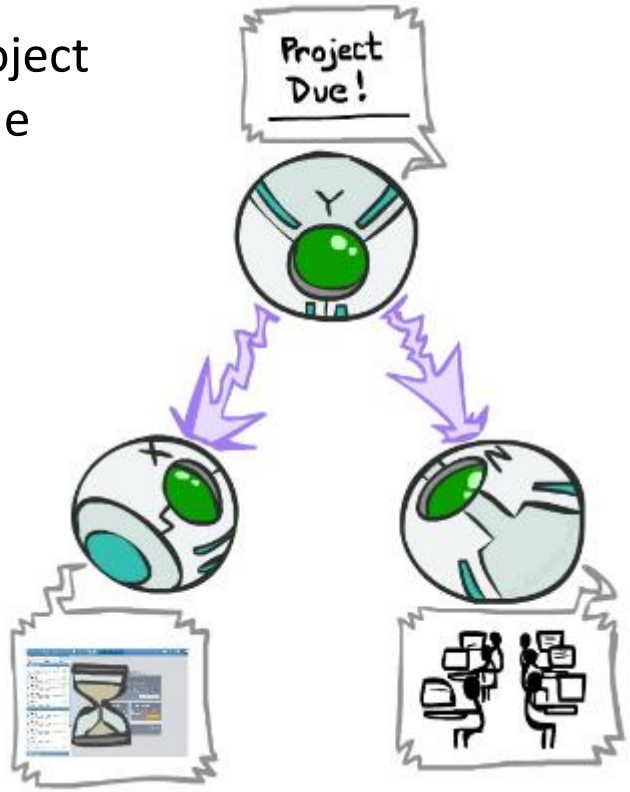
Yes!

- Evidence along the chain "blocks" the influence

Common Cause ("Confounder")

- This configuration is a "common cause"

Y: Project due



X: Forums busy

Z: Lab full

$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

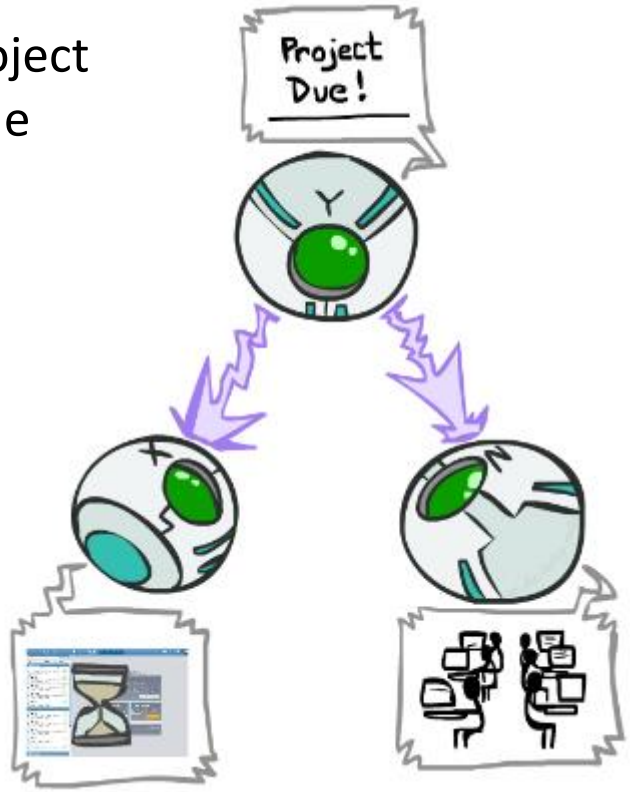
- Guaranteed X independent of Z ? *No!*
  - One example set of CPTs for which X is not independent of Z is sufficient to show this independence is not guaranteed.
  - Example:
    - Project due causes both forums busy and lab full
    - In numbers:

      P(+x | +y) = 1, P(-x | -y) = 1,
      P(+z | +y) = 1, P(-z | -y) = 1

Common Cause

- This configuration is a "common cause"

Y: Project due



X: Forums busy

Z: Lab full

$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

- Guaranteed X and Z independent given Y?

$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$

$$= \frac{P(y)P(x|y)P(z|y)}{P(y)P(x|y)}$$

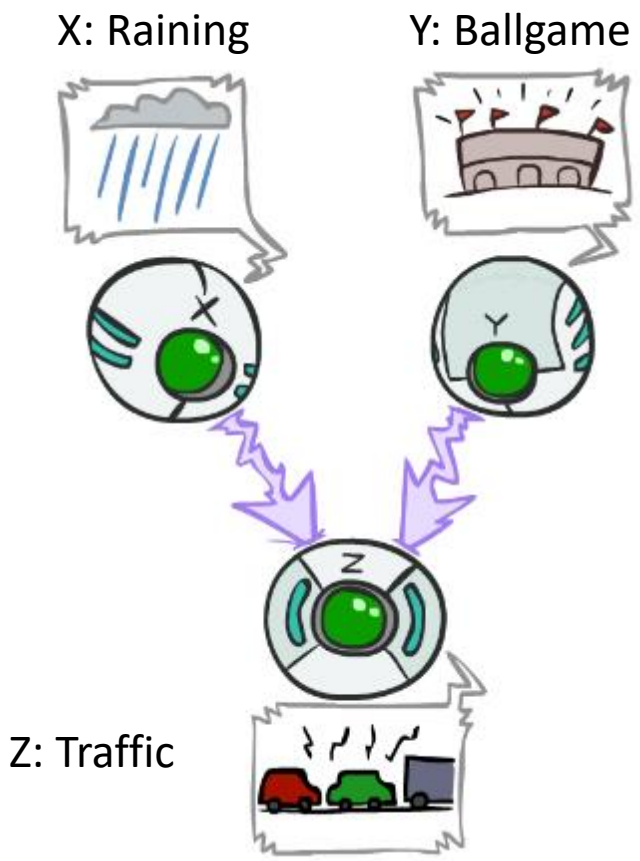
$$= P(z|y)$$

Yes!

- Observing the cause blocks influence between effects.

Common Effect ("Collider")

- Last configuration: two causes of one effect (v-structures)



- Are X and Y independent?
  - *Yes*: the ballgame and the rain cause traffic, but they are not correlated
  - Still need to prove they must be (try it!)
- Are X and Y independent given Z?
  - *No*: seeing traffic puts the rain and the ballgame in competition as explanation.
- This is backwards from the other cases
  - Observing an effect activates influence between possible causes.

The General Case

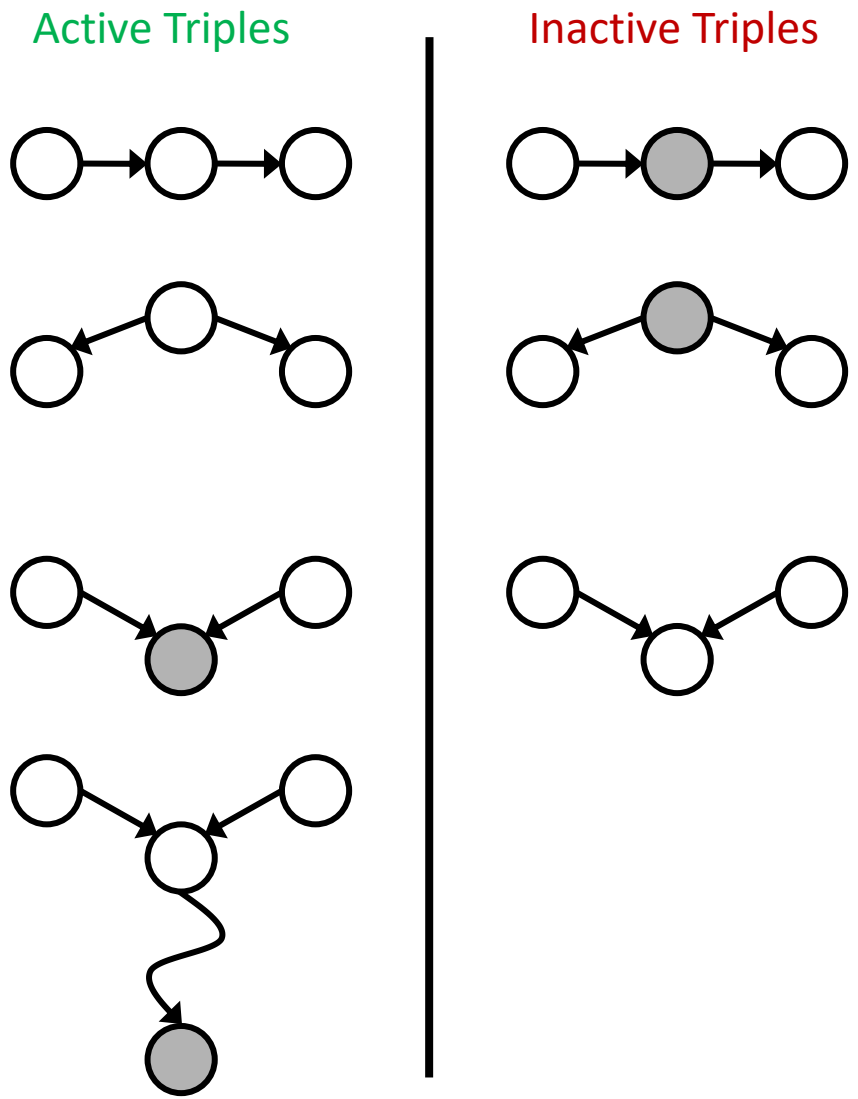


Active / Inactive Paths

- Question: Are X and Y conditionally independent given evidence variables {Z}?
  - Yes, if X and Y "d-separated" by Z
  - Consider all (undirected) paths from X to Y
  - No active paths = independence!

- A path is active if each triple is active:
  - Causal chain $A \rightarrow B \rightarrow C$ where B is unobserved (either direction)
  - Common cause $A \leftarrow B \rightarrow C$ where B is unobserved
  - Common effect (aka v-structure)
    $A \rightarrow B \leftarrow C$ where B *or one of its descendents* is observed

- All it takes to block a path is a single inactive segment



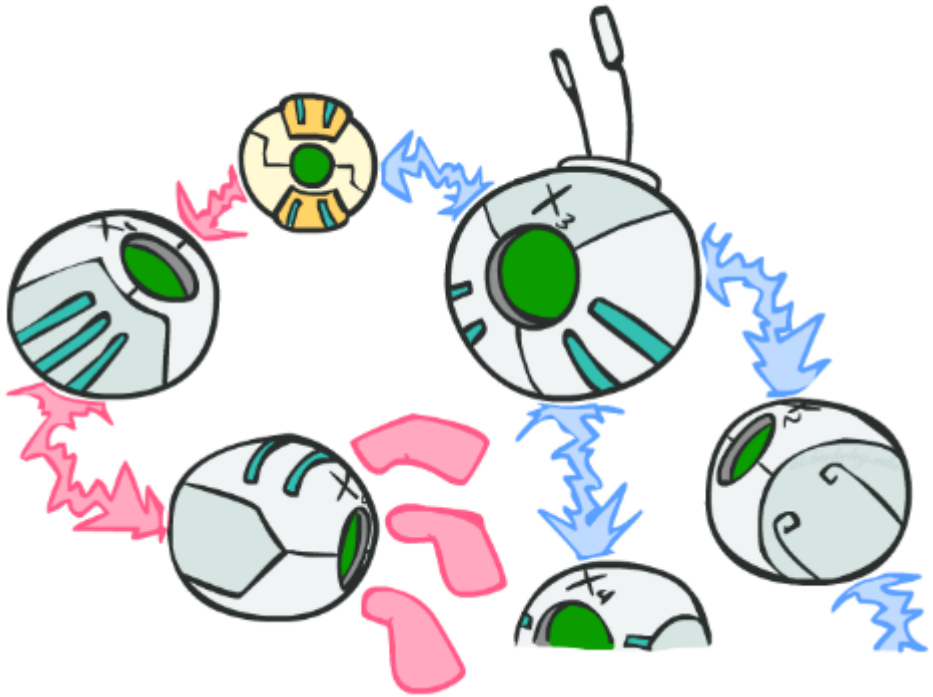
D-Separation

- Query: $X_i \perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$?
- Check all (undirected!) paths between X_i and X_j
  - If one or more active, then independence not guaranteed

$$X_i \not\perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$$

  - Otherwise (i.e. if all paths are inactive), then independence is guaranteed

$$X_i \perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$$

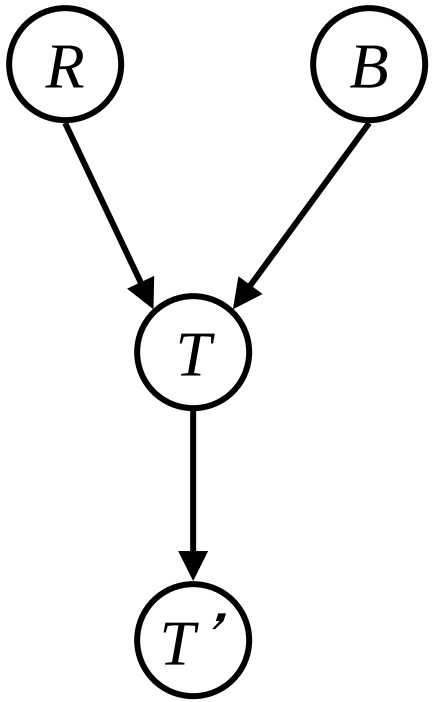


Example

$R \perp\!\!\!\perp B$ *Yes*

$R \perp\!\!\!\perp B | T$

$R \perp\!\!\!\perp B | T'$



Example

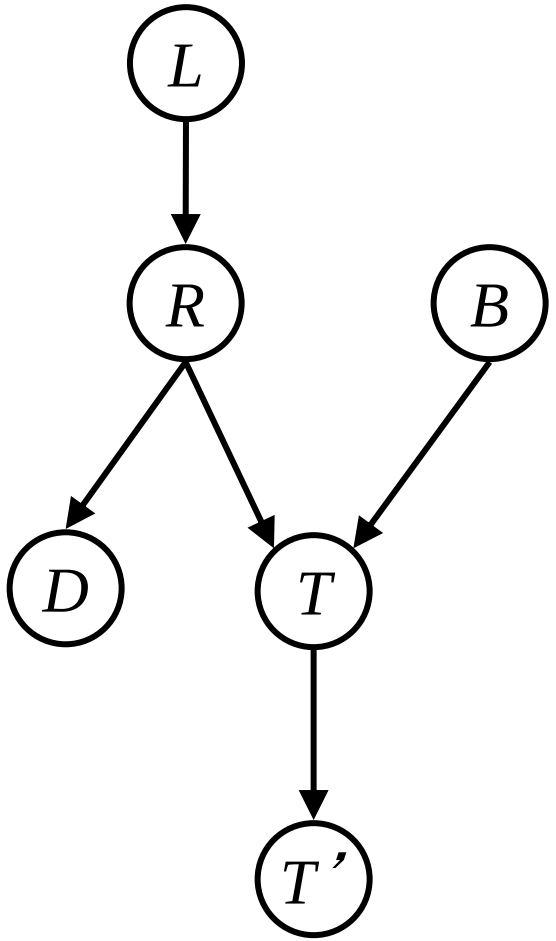
$L \perp\!\!\!\perp T' | T$ *Yes*

$L \perp\!\!\!\perp B$ *Yes*

$L \perp\!\!\!\perp B | T$

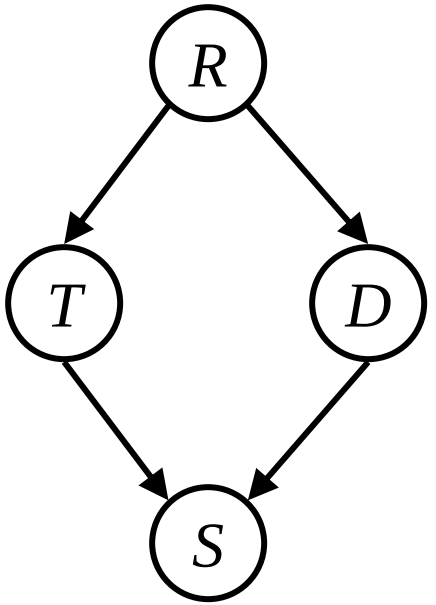
$L \perp\!\!\!\perp B | T'$

$L \perp\!\!\!\perp B | T, R$ *Yes*



Example

- Variables:
  - R: Raining
  - T: Traffic
  - D: Roof drips
  - S: I'm sad
- Questions:



$$T \perp\!\!\!\perp D$$

$$T \perp\!\!\!\perp D | R \qquad \textit{Yes}$$

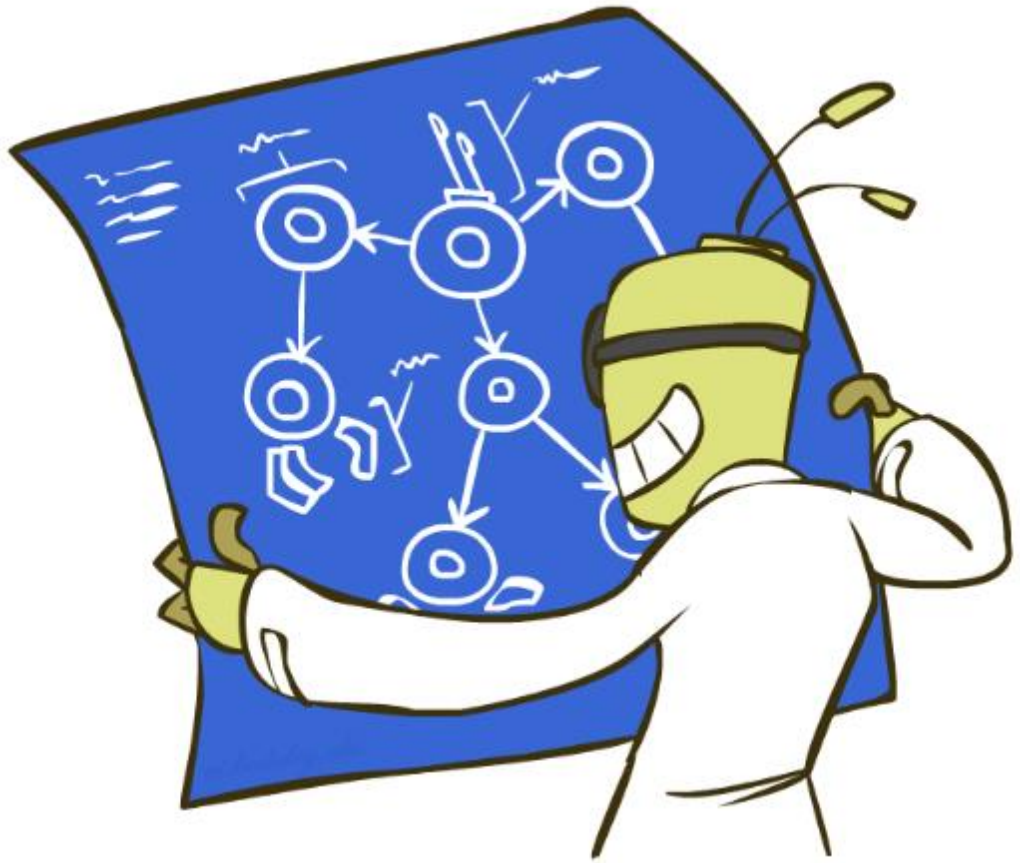
$$T \perp\!\!\!\perp D | R, S$$

Structure Implications

- Given a Bayes net structure, can run d-separation algorithm to build a complete list of conditional independences that are necessarily true of the form

$$X_i \perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$$

- This list determines the set of probability distributions that can be represented



Bayes Nets Representation Summary

- Bayes nets compactly encode joint distributions
- Guaranteed independencies of distributions can be deduced from BN graph structure
- D-separation gives precise conditional independence guarantees from graph alone
- A Bayes' net's joint distribution may have further (conditional) independence that is not detectable until you inspect its specific distribution

Motivating Example

- *Our patient is a known Asthma patient, has an elevated blood cell count, an elevated level of amylase, but a normal body temperature. What is the probability that our patient has a systemic inflammation reaction?*

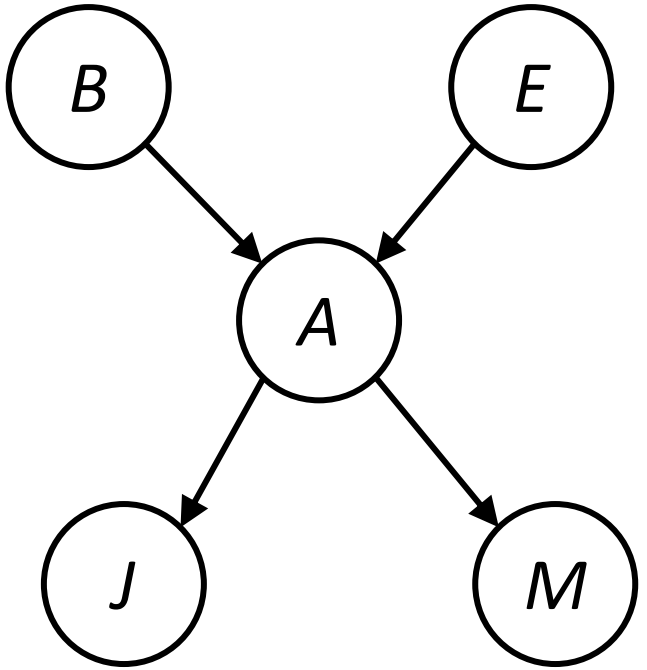
- How to model our knowledge?
- How to make inferences?

Next question we will concentrate on



Inference by Enumeration in Bayes' Net

- Given unlimited time, inference in BNs is easy
- Reminder of inference by enumeration by example:



$$P(B \mid +j, +m) \propto_B P(B, +j, +m)$$

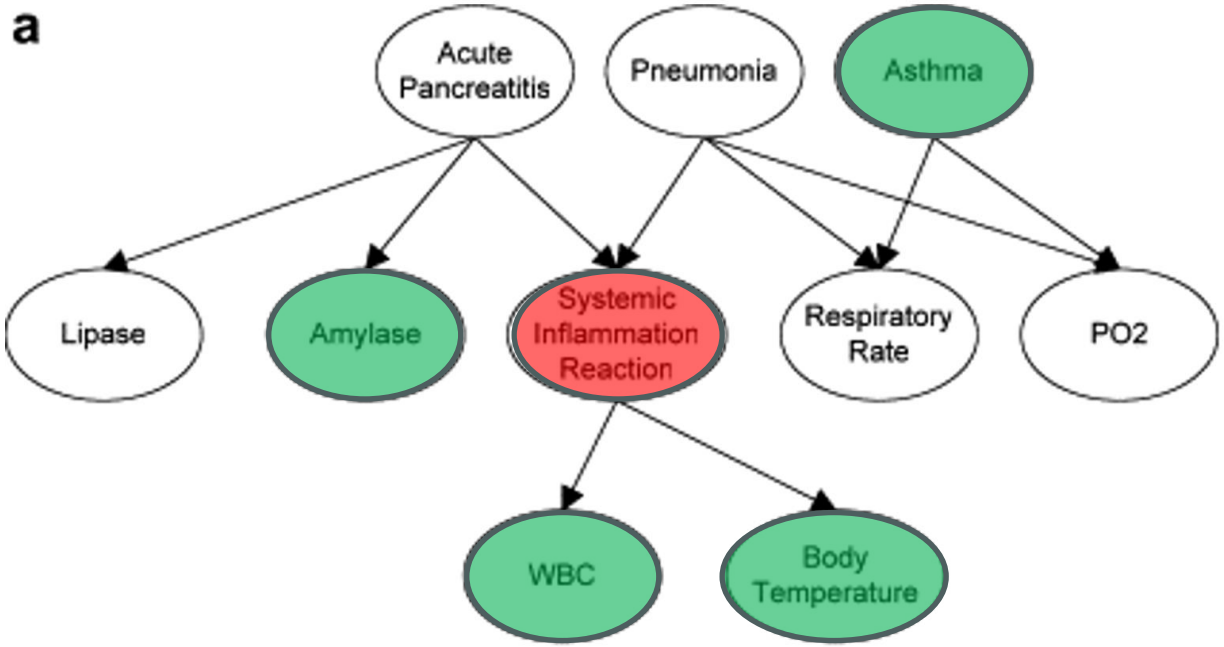
$$= \sum_{e,a} P(B, e, a, +j, +m)$$

$$= \sum_{e,a} P(B)P(e)P(a|B,e)P(+j|a)P(+m|a)$$

$$= P(B)P(+e)P(+a|B,+e)P(+j|+a)P(+m|+a) + P(B)P(+e)P(-a|B,+e)P(+j|-a)P(+m|-a)$$
$$P(B)P(-e)P(+a|B,-e)P(+j|+a)P(+m|+a) + P(B)P(-e)P(-a|B,-e)P(+j|-a)P(+m|-a)$$

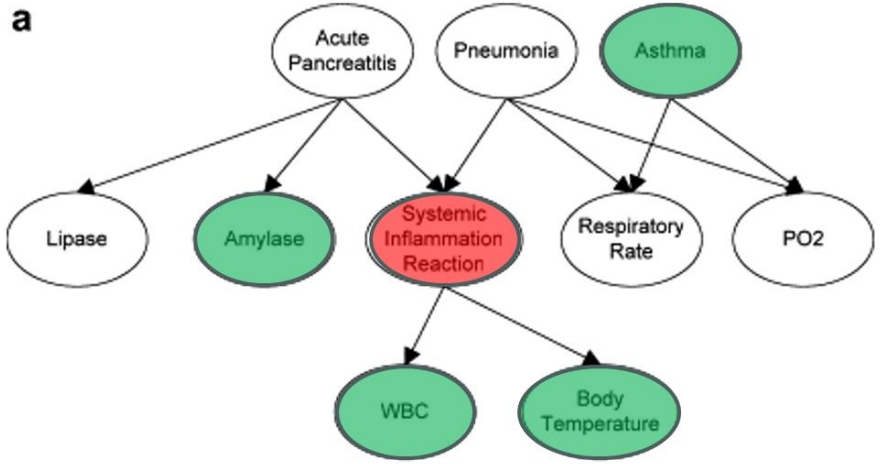
Making inferences

Our patient is a known Asthma patient, has an elevated blood cell count, an elevated level of amylase, but a normal body temperature. What is the probability that our patient has a systemic inflammation reaction?



$$P(S \mid Am, As, W, B)$$

Making inferences



$$P(S|As, Am, W, B)$$

$$= \frac{P(S, As, Am, W, B)}{P(As, Am, W, B)}$$

$$\propto P(S, As, Am, W, B)$$

$$= \sum_{ac}\sum_{pn}\sum_{l}\sum_{r}\sum_{po} P(ac, pn, l, r, po, S, As, Am, W, B)$$

$$= \sum_{ac}\sum_{pn}\sum_{l}\sum_{r}\sum_{po} P(ac)P(pn)P(As)P(l|ac)P(Am|ac)P(S|ac, pn)$$

$$P(r|pn, As)P(po|pn, As)P(W|S)P(B|S)$$

Making Inferences: Variable Elimination

- Reordering terms (using distributivity) increases efficiency:

$$\sum_e \sum_a P(B)P(e)P(a|B,e)P(+j|a)P(+m|a)$$

$$= P(B)\sum_e P(e)\sum_a P(a|B,e)P(+j|a)P(+m|a)$$

$$\sum_{ac}\sum_{pn}\sum_{l}\sum_{r}\sum_{po} P(ac)P(pn)P(As)P(l|ac)P(Am|ac)P(S|ac,pn)$$
$$\times P(r|pn,As)P(po|pn,As)P(W|S)P(B|S)$$

$$= P(W|S)P(B|S)P(As)\sum_{ac} P(ac)P(Am|ac)\sum_{pn} P(pn)\sum_{l} P(l|ac)P(S|ac,pn)$$
$$\times \sum_{r} P(r|pn,As)\sum_{po} P(po|pn,As)$$